

Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

MINI REVIEW 01 — Distance and Speed

Optional. Not marked by me, not recorded anywhere. We mark it together next week.

Where this fits the January mock. The mock covers Forces, Energy, Electricity, Particle and Atomic. This is the **first slice of Forces**. Packets 1–14 walk forward through Year 10 in taught order, so by the winter break you hold the whole of it in one folder.

Key concept

DISTANCE vs DISPLACEMENT (the difference is direction)

- **Distance** — how far you moved, added up along the whole path. No direction. **Scalar**.
- **Displacement** — the straight line from start to finish, *and* the direction of that line. **Vector**.

Walk 400 m right round a running track and stop where you started: distance **400 m**, displacement **0**.

Key concept

SPEED (scalar — no direction)

$$\text{distance} = \text{speed} \times \text{time} \quad s = vt$$

s in metres (m) · v in metres per second (m/s) · t in seconds (s)

Real speed is hardly ever constant — walking, running and driving all change speed constantly. When it does change, $s = vt$ with the **average** speed still works over the whole journey.

Typical speeds you are expected to recall:

walking	~1.5 m/s	running	~3 m/s
cycling	~6 m/s	sound in air	~330 m/s

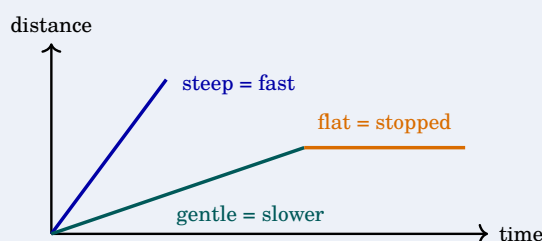
Worked example. A cyclist travels 1500 m in 250 s. Find the average speed.

Rearrange $s = vt$ to $v = \frac{s}{t} = \frac{1500}{250} = \mathbf{6 \text{ m/s}}$ — which is the typical cycling speed above, so the answer is believable. Always ask whether your number is a sensible speed for the thing that is moving.

Key concept

DISTANCE–TIME GRAPHS (distance up, time across)

The gradient is the speed. Steeper line, faster object.



A **flat** line means the object is **not moving** — time passes, distance does not change. It does *not* mean it has gone back to the start.

MINI REVIEW 01 — the core

Ten items, about ten minutes. Write something for each one even if you are not sure.

1. State whether each quantity is a **scalar** or a **vector**.

distance *scalar* displacement *vector* speed *scalar*

2. A runner completes one full lap of a 400 m track.

Distance travelled: *400 m* Displacement: *0 (she finishes where she started)*

3. Write down the equation linking distance, speed and time, in symbols.

$s = vt$ — distance = speed $\times$ time.

4. A car travels at a steady 25 m/s for 40 s. How far does it go?

$s = vt = 25 \times 40 = 1000 \text{ m}$ (1 km).

5. A walker covers 900 m in 600 s. Calculate the average speed.

$v = s/t = 900/600 = 1.5 \text{ m/s}$ — the typical walking speed.

6. Give the typical speed of each, in m/s.

walking *1.5 m/s* running *3 m/s*

cycling *6 m/s* sound in air *330 m/s*

7. On a distance–time graph, what does the **gradient** of the line tell you?

The speed of the object.

8. On a distance–time graph, what does a **horizontal (flat)** line mean?

The object is stationary — time is passing but the distance is not changing.

9. Two straight lines are drawn on the same distance–time graph. Line A is steeper than line B. Which object is moving faster, and how do you know?

A. A steeper gradient means more distance covered in the same time, and the gradient of a distance–time graph is the speed.

10. Thunder is heard 6 s after the lightning is seen. Taking the speed of sound in air as 330 m/s, how far away was the strike?

$s = vt = 330 \times 6 = 1980 \text{ m}$, about 2 km. The light arrives effectively instantly, so the 6 s is all sound travel time.

Stretch — not marked in class. A student walks 30 m north, then 40 m east, in a total of 50 s. (a) What total *distance* did she walk? (b) What was her *displacement*? (c) What was her average *speed*?

(a) $30 + 40 = 70 \text{ m}$. (b) The path is a right angle, so the straight line from start to finish is $\sqrt{30^2 + 40^2} = 50 \text{ m}$, in a north-easterly direction — and the direction is part of the answer. (c) Average speed uses distance: $70/50 = 1.4 \text{ m/s}$, not $50/50$ — that would be average velocity, which is R02.